

Week 2 – Security Planning & Testing Methodology

2.1 – Performance Testing Plan

Performance testing is conducted remotely from the workstation using SSH. Baseline measurements are recorded on an idle system, followed by controlled workload execution. Metrics are collected during CPU, memory, disk, and network stress to evaluate operating system performance under load

2.2 Security Configuration Checklist

- SSH hardening (key-based authentication)
- Firewall restricting SSH access to a single workstation IP
- Mandatory Access Control
- Automatic security updates
- User privilege management
- Network monitoring and logging

2.3 Threat Model

Threat	Description	Mitigation
Malware	Exploits vulnerabilities to gain control	Automatic updates, limited sudo access
DoS Attack	Resource exhaustion	Firewall rules, monitoring
SSH Brute Force	Credential guessing	SSH keys, fail2ban, login attempt limits